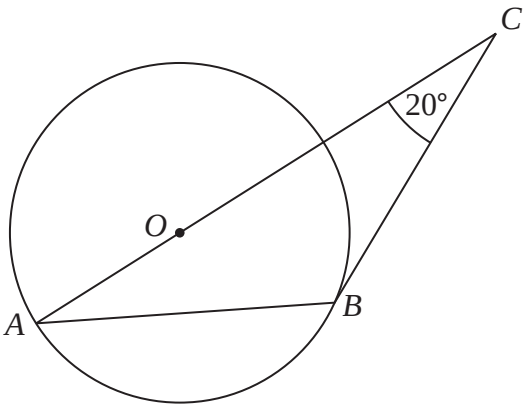
13



A and *B* are points on the circumference of a circle, centre *O*.
BC is the tangent to the circle at *B*.
AOC is a straight line.

Angle *ACB* = 20°

Find the size of angle *CBA*.
You must show all your working.

Angle OBC = 90° (radius meets tangent)

1 mark
Radius perpendicular to tangent.

Angle BOC = 180° – 90° – 20° = 70°

Angle AOB = 180° – 70° = 110°

OA = OB, so angle ABO = (180° –110°) ÷2

= 35°

1 mark
AOC is straight;
OA = OB, so equal base angles.

Angle CBA = 90° +35° = 125°

125 °

(Total for Question 13 is 3 marks)

1 mark
Add angles CBO and OBA to find the required whole angle.

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